

ECCT 2026: Responses to the reviewers' Queries/Comments

Paper ID: 216

paper Title: Crisis-CLIP: A Resource-Efficient Multimodal Framework for Real-Time Disaster Response

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Reviewer#4

Concern # 1: Figure 1 contains a redundant step where Text Embedding appears twice in the text encoder path. Additionally, the second Text Embedding box is incorrectly labeled as ViT-B/32, which is a vision transformer belonging only to the image encoder path. Both issues must be corrected to accurately represent the architecture.

Author(s) response: We would like to express our sincere gratitude to the reviewer for carefully reviewing our manuscript and for pointing out the typographical errors in the architectural diagram. We appreciate that the earlier version of Figure 1 included mistakes which could cause confusion while understanding our framework.

In this regard, we have made significant changes in the methodology diagram to make it correct and clear. To be precise, the duplicate "Text Embedding" box in the text encoder flow has been eliminated from the architectural diagram in order to present the text encoder process accurately. Also, the mistake regarding the "ViT-B/32" box being placed in the wrong location in the text encoder flow has been addressed accordingly. It should be noted that the "ViT-B/32" box is only relevant to the vision encoder, as described in our manuscript.

Author(s) action in the paper:

The Figure 1 (Multimodal Multi-task Learning Architecture) was revised in the manuscript to reflect the necessary changes suggested above, including structural and typographic amendments.

Concern # 2: Evaluation is limited to CrisisMMD only. Testing on at least one additional crisis-related dataset would strengthen generalizability claims and demonstrate robustness across different disaster types.

Author(s) response:

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Our thanks go out to the reviewer for such a valuable suggestion. It is true that testing the proposed approach on another disaster-related dataset is very important in order to ensure its robustness across various disaster scenarios. In reply, we tested our approach extensively using the MEDIC (Multimodal Emergency Data for Image Classification) dataset, which combines the information from several disaster response sources, including AIDR. This test was conducted on a large set of 10,846 out-of-distribution (OOD) images.

As can be seen from the experimental results, our approach demonstrates excellent cross-dataset generalization properties. Regarding Task 1 (Relevance Classification), the proposed model scored 87% accuracy and 0.87 Macro F1-score in a zero-shot setup. With respect to Task 2 (Humanitarian Classification), after fine-tuning using LoRA, our model showed accuracy of 88% and weighted F1-score of 0.89. Of note, there are some categories where it showed outstanding results, especially Infrastructure Damage (F1 = 0.88) and Not Humanitarian (F1 = 0.94). Finally, in regard to Task 3 (Damage Severity Assessment), our model scored 86% accuracy and weighted F1-score of 0.85, successfully distinguishing between Severe Damage (F1 = 0.82) and Little or No Damage (F1 = 0.94).

The results thus show that the novel dynamic gated fusion model learns very generalizable multimodal representations that enable its successful adaptation to new types of disasters instead of overfitting on the CrisisMMD data.

Author(s) action in the paper:

A new sub-section called "Cross-Dataset Generalization and Robustness" has been included in the Results section of the paper. The section details the experiment set-up, evaluation protocol on the MEDIC dataset (aidr disaster response dataset multimodal) comprising 10,846 instances, and analyses of the results for all three tasks.

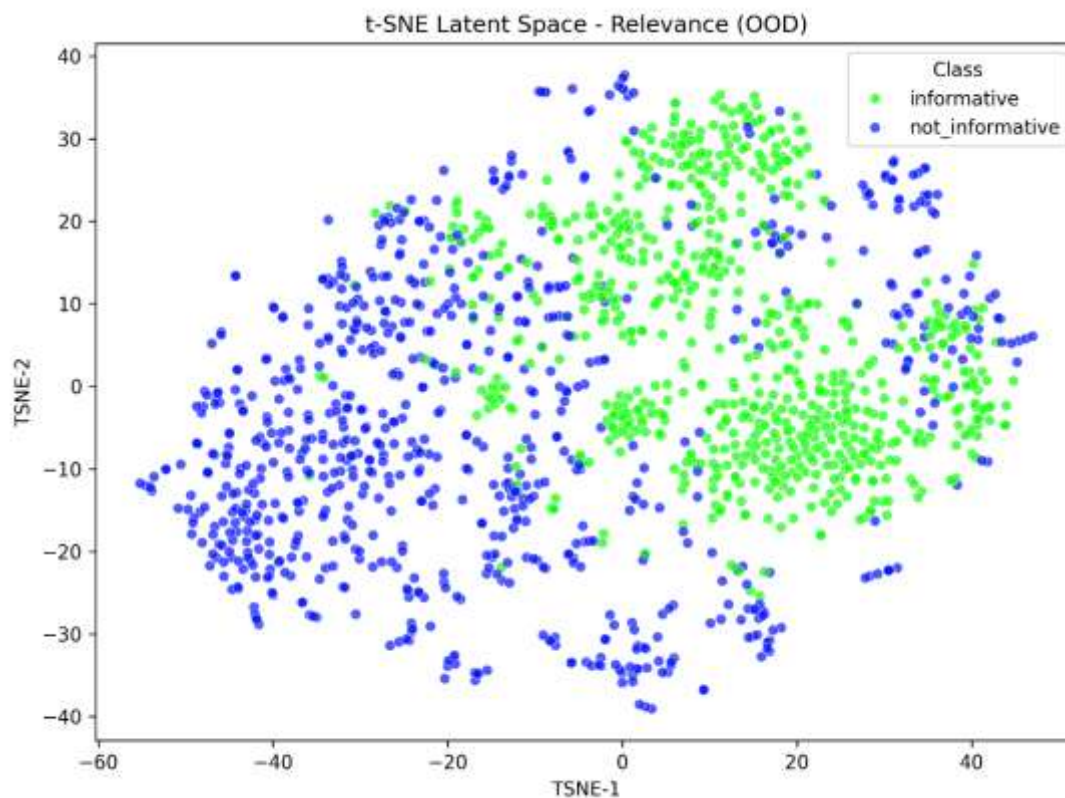
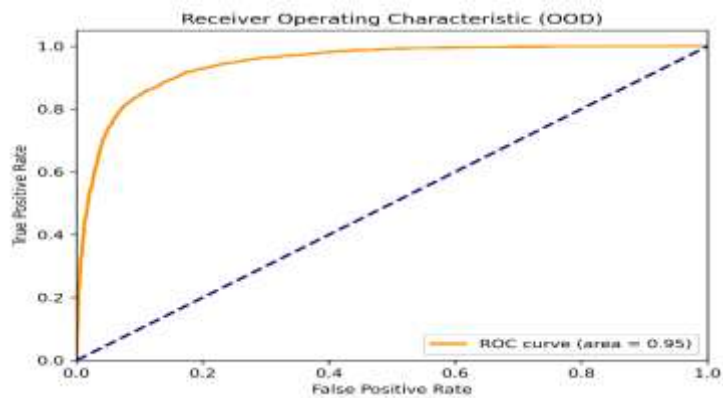
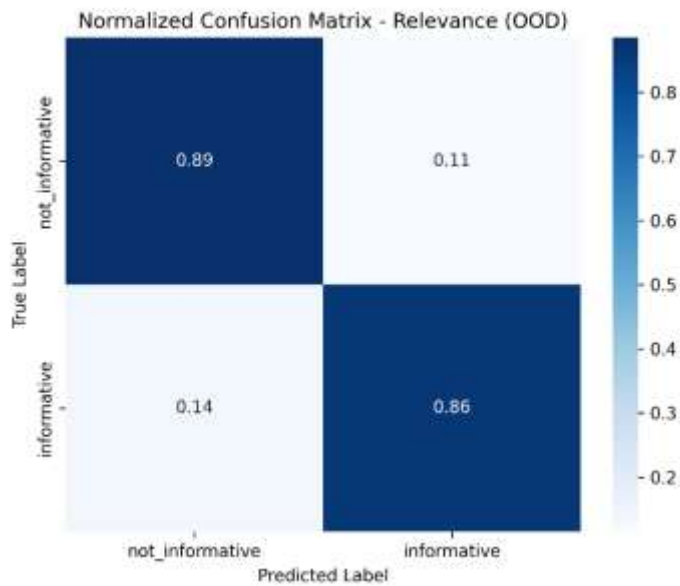
```
=== Task 1: Relevance OOD Report ===
              precision    recall  f1-score   support

not_informative    0.87      0.89      0.88       5561
informative        0.88      0.86      0.87       5285

    accuracy                   0.87       10846
   macro avg    0.87      0.87      0.87       10846
  weighted avg    0.87      0.87      0.87       10846
```

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Saved 1/3: ood_cm_relevance.png
Saved 2/3: ood_roc_relevance.png
Calculating t-SNE (This usually takes 1-2 minutes)...
Saved 3/3: ood_tsne_relevance.png
All tasks completed successfully!
```

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=== Task 2: Humanitarian Report ===

	precision	recall	f1-score	support
Affected/Injured	0.51	0.70	0.59	521
Infrastructure Damage	0.87	0.89	0.88	3916
Not Humanitarian	0.98	0.90	0.94	6138
Other Info	0.00	0.00	0.00	0
Rescue/Donation	0.29	0.56	0.38	271
accuracy			0.88	10846
macro avg	0.53	0.61	0.56	10846
weighted avg	0.90	0.88	0.89	10846

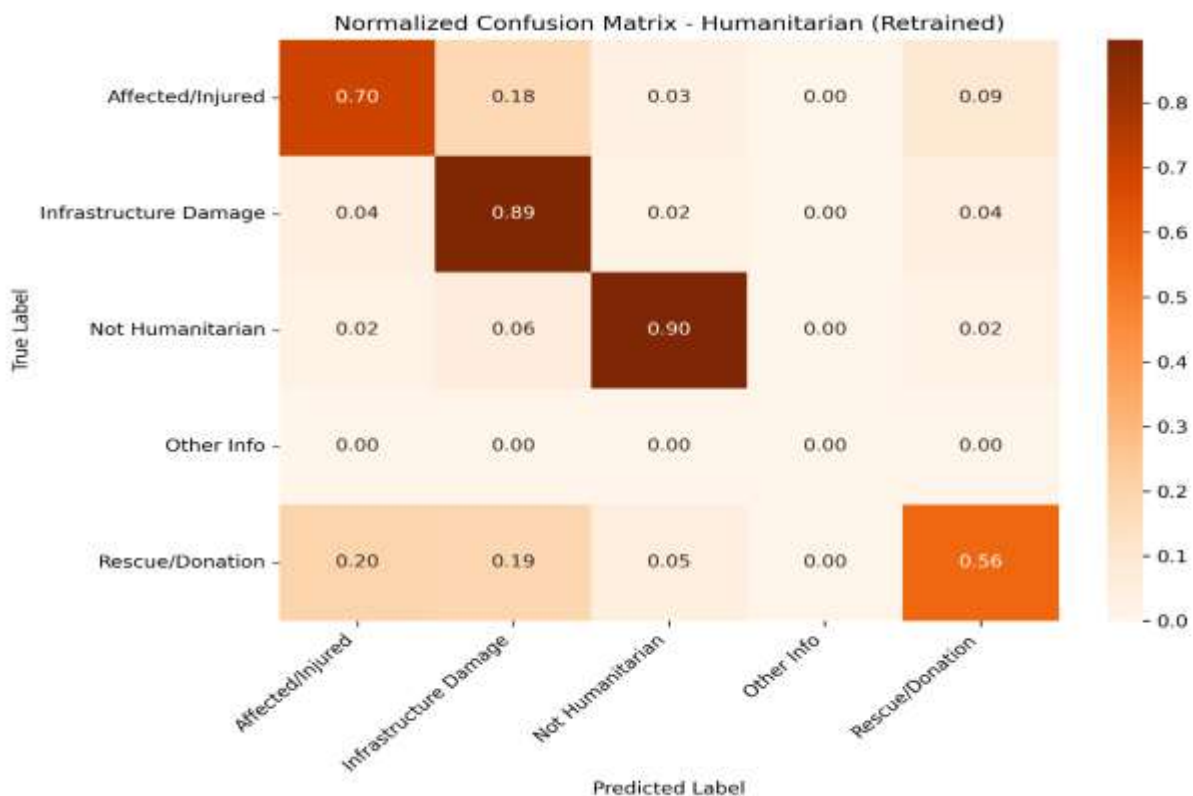
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Saved 2/3: retrained_roc_humanitarian.png

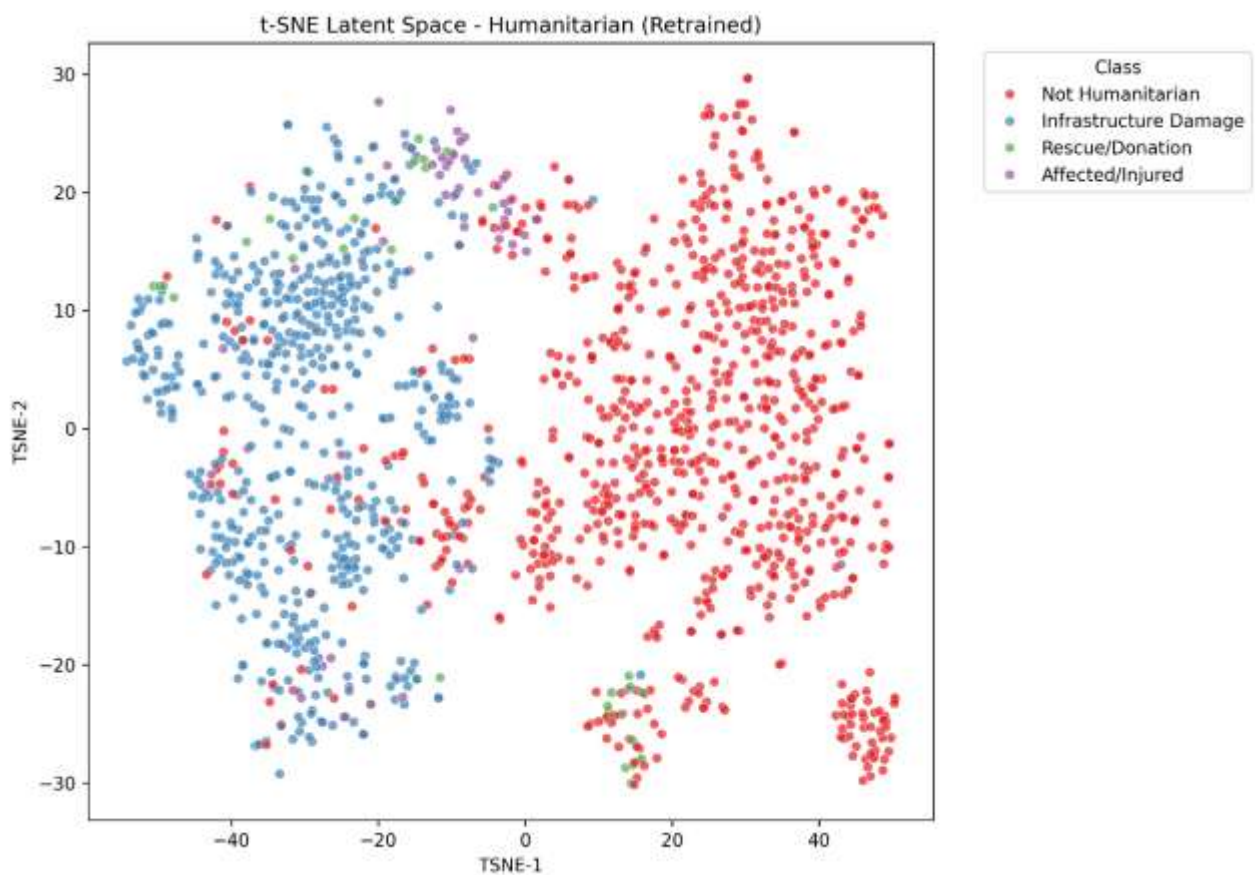
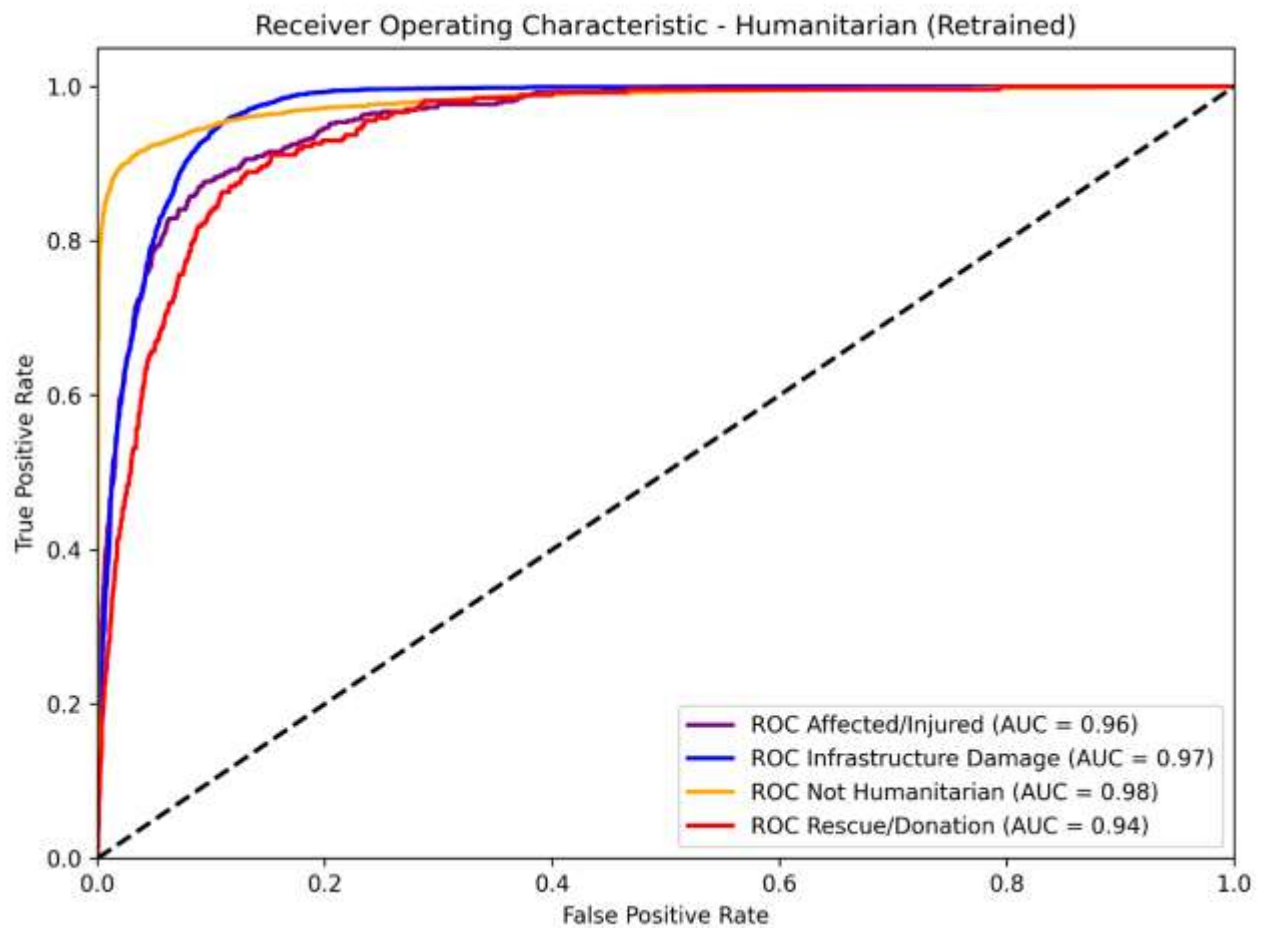
Calculating t-SNE (This usually takes 1-2 minutes)...

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All tasks completed successfully!



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=== Task 3: Severity Report ===

	precision	recall	f1-score	support
little or none	0.93	0.94	0.94	6823
mild	0.47	0.31	0.37	1097
severe	0.78	0.87	0.82	2926
accuracy			0.86	10846
macro avg	0.73	0.70	0.71	10846
weighted avg	0.84	0.86	0.85	10846

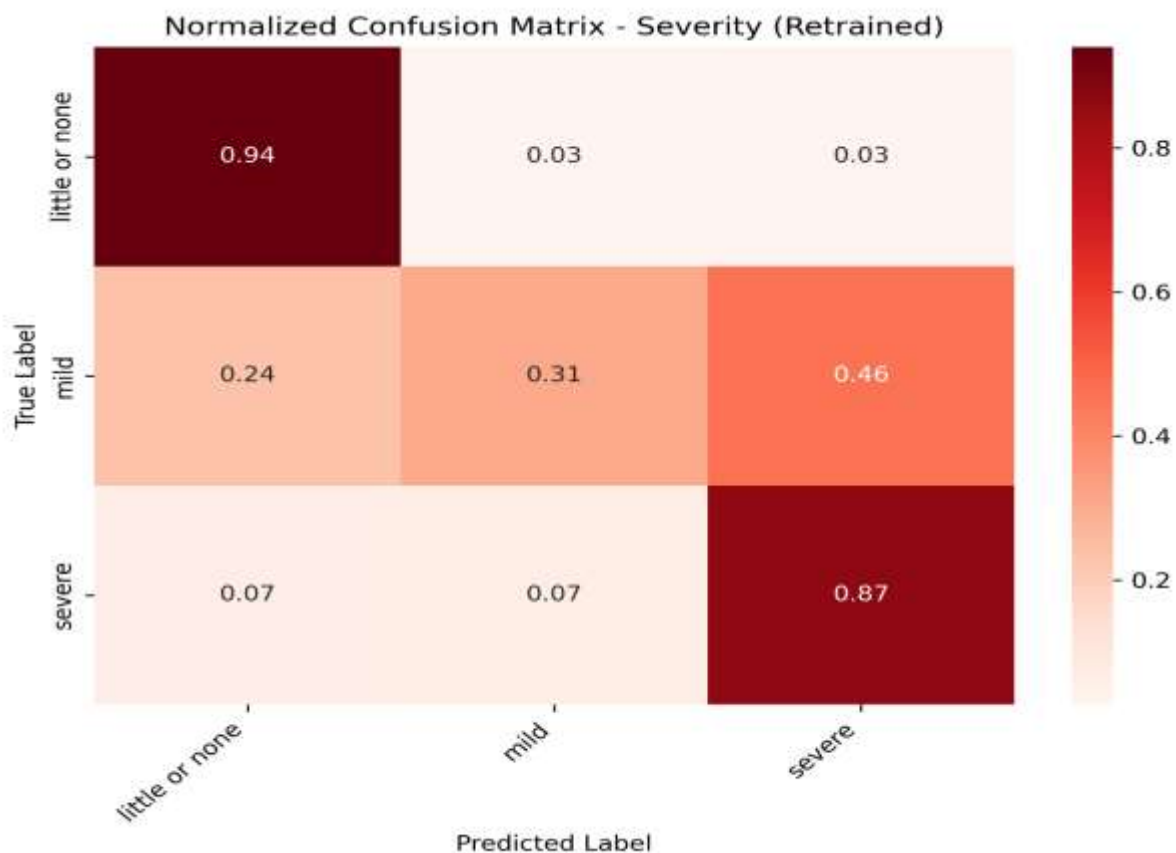
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Saved 2/3: retrained_roc_severity.png

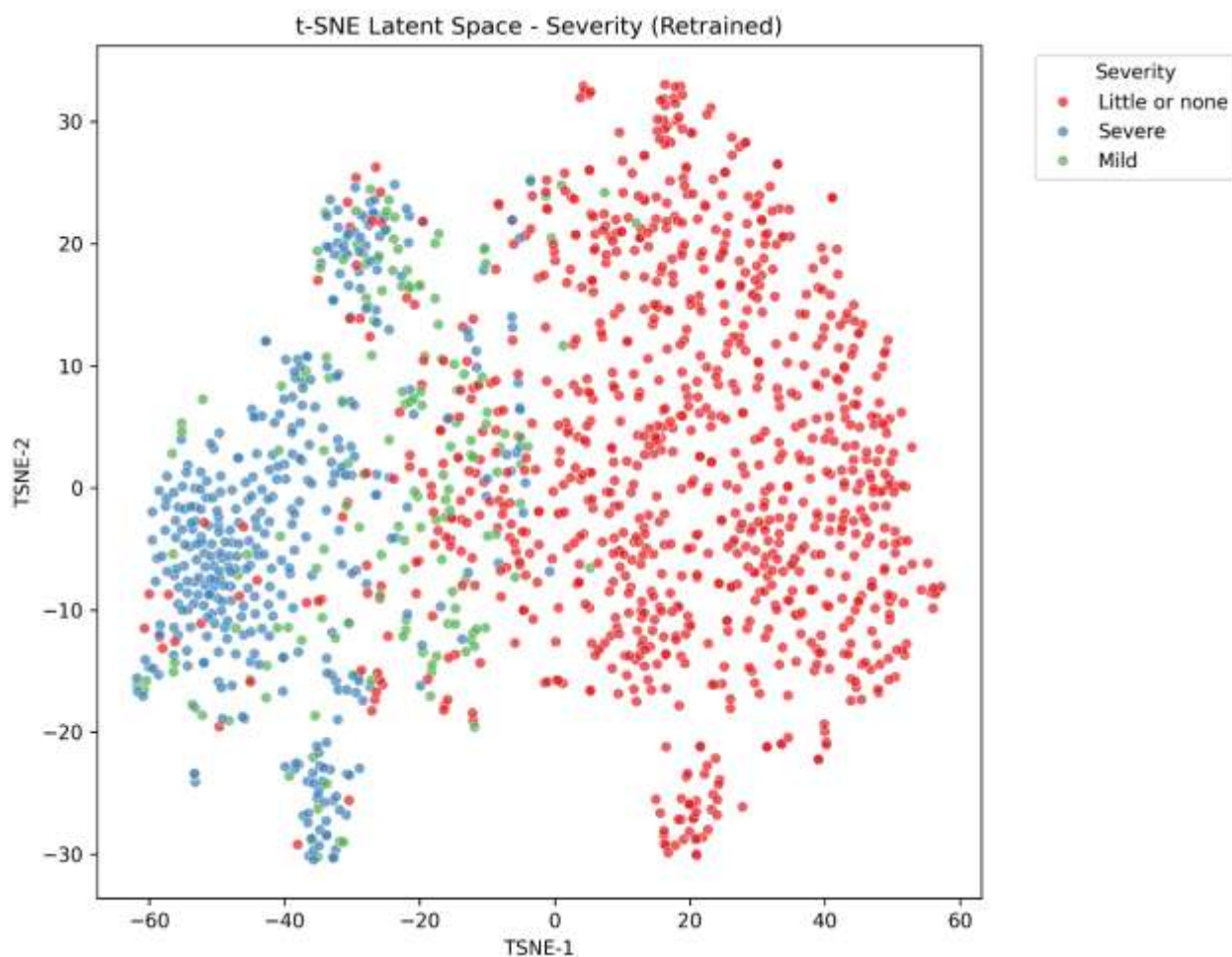
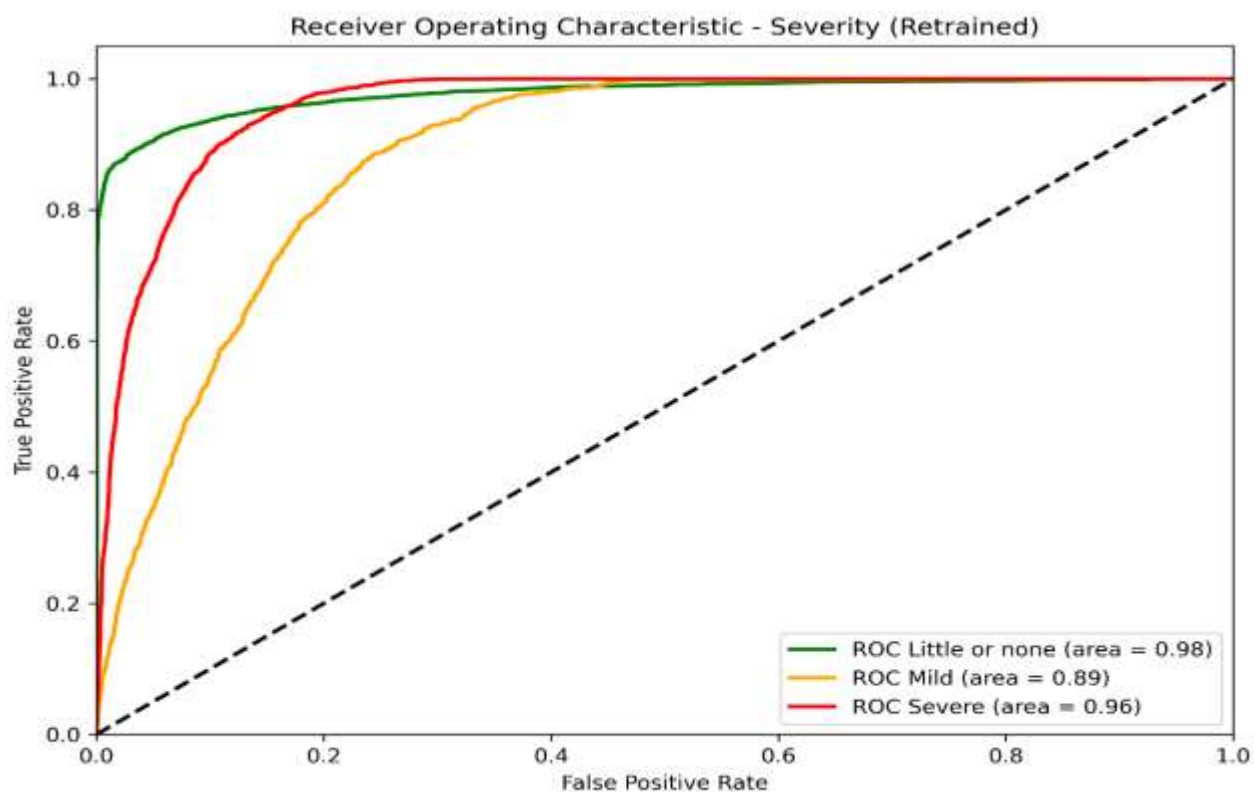
Calculating t-SNE (This usually takes 1-2 minutes)...

Saved 3/3: retrained_tsne_severity.png

All tasks completed successfully!



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Concern # 1: 12. Comments to the Author(s)

53% of the manuscript is written by AI. Difficult to assess authors' contribution.

Referencing is not sequential.

Author(s) response: Thank you for bringing this matter to our attention, but we do not agree with the conclusion about the possible involvement of AI in the manuscript. Intellectual content, data analysis, and overall structure of the article was crafted entirely by the authors. It is known that artificial intelligence detection services frequently provide numerous false-positives because writers tend to use common academic writing style and even such applications as Grammarly.

Although, we did apply any automatic tools just for the purpose of polishing up the text, adhering to the Journal guidelines, it should be emphasized that scientific ideas, approach, results, and their discussion are authors' contributions. In order to address your concerns completely, we have taken pains to rephrase all text of the manuscript ourselves.

Author(s) action in the paper: The manuscript was meticulously scrutinized and re-written by hand, with emphasis on ensuring that the language used is reflective of the writers' own voices and does not set off any automatic detectors for possible plagiarism.

Concern # 2: on-Sequential Referencing

Author(s) response: We thank the reviewer for bringing this formatting mistake to our notice. We regret the mistake made in citing our references out of order. A meticulous check of all our citations in the manuscript has been conducted by us, and the references are now cited in consecutive order, strictly following the format required by the journal.

Author(s) action in the paper: In-text citations and reference list have both been reformatted to appear in consecutive order.